

Comparative Study of SRGAN and SRRaGAN Architectures for Anime Image Super Resolution

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Abstract—Single-image super-resolution enhances low-resolution images for applications in medical imaging, video streaming, surveillance, and content restoration. In the anime industry, super-resolution is essential for upscaling legacy content and maintaining streaming quality. However, anime imagery presents unique challenges in preserving hand-drawn lines that are fundamental to anime aesthetics. Traditional interpolation methods fail to preserve these features, resulting in blurred edges and loss of fine details. This study compares four training strategies applied to a ResNet-based generator architecture for anime single-image super-resolution at 4× scaling: SRResNet, SRGAN, SRRaResNet, and SRRaGAN. A dataset of 24,340 safe-for-work anime images was collected from Danbooru 2024, which cover diverse art styles and character designs. SRResNet trains the ResNet generator with MSE loss as a baseline. SRGAN uses the same ResNet architecture as its generator, initialized from the pre-trained SRResNet checkpoint, combined with a discriminator network and perceptual loss based on VGG19 features for adversarial training. Both SRRaResNet and SRRaGAN uses a combination of perceptual and pixel-level L1 loss for enhanced detail preservation, with SRRaGAN employing relativistic adversarial training for improved stability. All models were evaluated using Peak Signal-to-Noise Ratio, Structural Similarity Index, and qualitative visual assessments focusing on line preservation and artistic fidelity. Results provide insights into how different training strategies and architectural modifications affect the ResNet generator’s performance for anime super-resolution, with implications for content restoration and streaming.

Index Terms—single-image super-resolution, generative adversarial network, anime imagery

I. INTRODUCTION

Image super-resolution is a computational technique that enhances the resolution of digital images by reconstructing high-quality versions from low-resolution inputs. This process involves predicting and generating missing pixel information to create sharper, more detailed images without introducing significant visual artifacts. Super-resolution technology is widely applied across domains including medical imaging for clearer diagnostic scans, video streaming platforms for adaptive quality improvement, surveillance systems for enhanced identification capabilities, and digital content restoration for preserving historical media.

The code is available at <https://github.com/sirreidlos/aol-dl>. The application is accessible at <https://erik-san.dev/aol-dl>.

In the entertainment and media industry, particularly in anime production and distribution, super-resolution plays a crucial role in upscaling legacy content for modern high-definition displays and maintaining consistent streaming quality across varying network conditions. The anime industry presents unique challenges for image enhancement due to its distinctive visual characteristics. Unlike photographic images, anime artwork features hand-drawn lines that serve as fundamental visual elements, requiring precise preservation during enhancement. These lines are often distorted and faint in low-resolution images, making their accurate reconstruction particularly challenging. Traditional upscaling methods such as bicubic and bilinear interpolation fail to preserve the clarity and sharpness of these critical features, resulting in blurred edges, loss of fine details, and overall visual degradation [1] [2] [3]. These limitations highlight the need for more sophisticated approaches capable of reconstructing anime-specific visual elements while maintaining artistic integrity.

This study compares four training approaches applied to the ResNet generator architecture for anime single-image super-resolution: SRResNet, SRGAN, SRRaResNet (ResNet without batch normalization trained by combining perceptual and pixel-level L1 loss), and SRRaGAN (initialized from pre-trained SRRaResNet, with relativistic adversarial training [4]). This design explores both the impact of architectural modifications (batch normalization) and training strategies (relative adversarial) on super-resolution quality. Models are evaluated using quantitative metrics (PSNR and SSIM) and qualitative assessments of visual and artistic quality. The goal is to determine how different training strategies and architectural choices affect the ResNet generator’s ability to enhance anime imagery, particularly in reconstructing the precise line work critical to anime aesthetics, providing insights for improving legacy content, streaming quality, and display optimization.

II. RELATED WORK

Deep learning has transformed single-image super-resolution by enabling algorithms to learn complex relationships between low-resolution and high-resolution image pairs [1]. Residual Networks (ResNets) [5] leverage skip connections and residual learning to facilitate deeper network architectures, enabling the model to learn fine-grained image

details through progressive refinement across multiple layers. This architecture addresses the vanishing gradient problem inherent in deep networks, allowing for more effective feature extraction and reconstruction. ResNet architecture can be trained using different strategies to optimize various objectives. Removing batch normalization has been found to help with stabilization [6], as batch statistics can sometimes introduce unpleasant artifacts. The ResNet architecture also functions as the generator in Generative Adversarial Networks (GANs). Super-Resolution Generative Adversarial Networks (SRGAN) [2] employ this ResNet generator, typically initialized from a pre-trained SRResNet checkpoint, paired with a discriminator network in an adversarial training framework, utilizing perceptual loss functions based on VGG19 [7] feature representations [8] to produce photo-realistic images with sharper details and enhanced visual fidelity. Recent advances have introduced relativistic adversarial training [4][6], which compares the realism of generated images relative to real images rather than in isolation, leading to more stable training and improved perceptual quality.

III. DATASET

The dataset utilized in this study was constructed through systematic scraping of the Danbooru 2024 metadata repository, accessible via the deepghs/danbooru2024 collection on Hugging Face [9]. To ensure content appropriateness, we applied filtering criteria to retain only safe-for-work (SFW) rated images, yielding a final dataset comprising approximately 24,340 high-resolution images. The dataset represents an aggregate collection of anime-style artwork and fan-created illustrations sourced from various artists, encompassing diverse art styles, character designs, compositional structures, and color palettes.

The characteristic of this dataset is its unfiltered nature, with no manual quality control implemented. This approach presents both advantages and limitations. The primary strength lies in the dataset's raw diversity, exposing the model to a wide spectrum of artistic variations, quality levels, and stylistic expressions that reflect real-world artistic production. But at the same time, the absence of quality curation may introduce inconsistencies in image quality and artistic coherence.

IV. MODEL ARCHITECTURE AND TRAINING

The initial experimental design compared SRResNet and SRGAN to highlight the performance difference between MSE-based super-resolution and adversarial super-resolution. However, repeated training trials showed that SRGAN frequently experienced instability and mode collapse, preventing consistent convergence.

A review of related research identified ESRGAN [6] as a more stable adversarial approach with improved perceptual quality. However, ESRGAN introduces substantial architectural complexity that did not align with the resource limitations and comparative scope of this study.

To achieve improved training stability without excessive architectural overhead, specific ESRGAN enhancements were selectively integrated into the SRGAN baseline rather than

adopting the full ESRGAN architecture. The modifications included the removal of batch normalization layers from the generator and the implementation of relativistic adversarial loss, resulting in the SRRaResNet and SRRaGAN models. This approach preserves the comparative simplicity of the original SRGAN framework while benefiting from ESRGAN's stability improvements.

Preliminary trials were conducted that included batch normalization layers in both SRRaResNet and SRRaGAN. These early configurations consistently suffered from training instability and rapid collapse once visual artifacts emerged. Based on these observations and supported by prior findings in the literature, batch normalization layers were removed in the final experimental design to improve stability and visual consistency.

This study implements a ResNet-based generator architecture with two architectural variants trained under four different strategies. Two strategies (SRGAN and SRRaGAN) incorporate an additional discriminator network for adversarial training. The key architectural difference between model pairs lies in the inclusion or exclusion of batch normalization layers.

A. Generator

The generator architecture in Fig. 2 employs a deep residual learning framework consisting of three main components, feature extraction, residual learning blocks, and upscaling modules. The network for feature extraction begins with a single 9×9 convolutional layer that projects the input low-resolution image (3 channels) into a 64-channel feature space, followed by a PReLU activation function. The core comprises 16 residual blocks (N=16), each implementing skip connections to enable efficient gradient flow and learning of residual mappings. Each residual block contains two 3×3 convolutional layers with PReLU activations. Two architectural variants exist (Fig. 1):

- With Batch Normalization: Each convolutional layer is followed by batch normalization before PReLU activation.
- Without Batch Normalization [6]: Batch normalization layers are removed to preserve fine details and sharp edges.

Following the residual blocks, another 3×3 convolutional layer is applied, and a global skip connection from the initial feature extraction layer is added to preserve low-level features throughout the deep architecture.

The upsampling stage utilizes sub-pixel convolutional blocks (M=2 blocks for 4× scaling) that progressively increase spatial resolution. Each block applies a 3×3 convolution to expand channels by a factor of 4 (from 64 to 256 channels), followed by pixel shuffle operation to rearrange these channels into spatial dimensions (increasing resolution by 2×), and PReLU activation. Finally, a 9×9 convolutional layer reconstructs the 3-channel RGB output image.

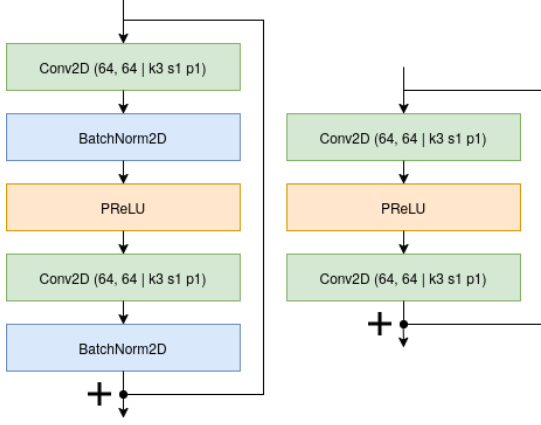


Fig. 1: ResidualBlock

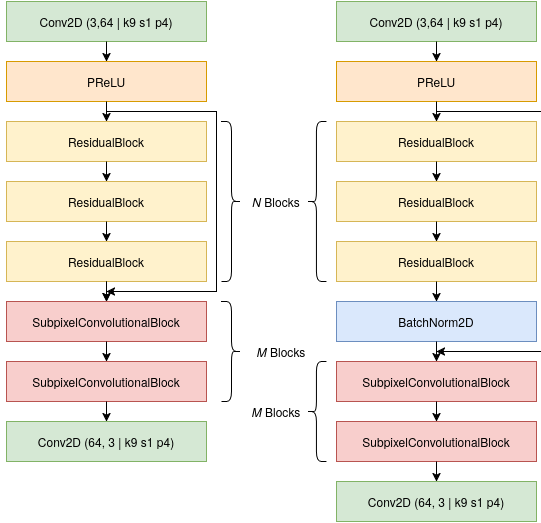


Fig. 2: Generator Architecture

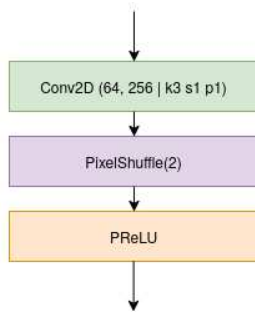


Fig. 3: Subpixel Convolutional Block (Upsampling)

B. Discriminator

The discriminator network, shared by both SRGAN and SRRaGAN distinguishes between generated super-resolution images and authentic high-resolution images through a series of convolutional blocks (Fig. 4).

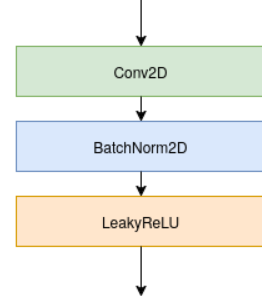


Fig. 4: Discriminator Convolutional Block

The architecture, as shown in Fig. 5, begins with an initial 3×3 convolutional layer that maps the input RGB image to 64 channels, followed by LeakyReLU activation (negative slope 0.2). Seven convolutional blocks then progressively extract deeper features while reducing spatial dimensions. Each convolutional block consists of a 3×3 convolution, batch normalization, and LeakyReLU activation. The blocks alternate between:

- Even indices (0, 2, 4, 6): Spatial downsampling with stride 2, maintaining channel count.
- Odd indices (1, 3, 5): Channel expansion (doubling channels), stride 1.

This progression produces feature maps with increasing depth: $64 \rightarrow 64 \rightarrow 128 \rightarrow 128 \rightarrow 256 \rightarrow 256 \rightarrow 512 \rightarrow 512$ channels.

The deep feature representation is then processed through adaptive average pooling to a fixed 6×6 spatial size, flattened, and passed through two fully connected layers ($512 \times 36 \rightarrow 1024 \rightarrow 1$) with LeakyReLU activation, outputting a single logit representing the probability of the input being real.

C. Training Strategy

1) *SRResNet*: SRResNet serves as the baseline training approach, where the generator is trained using only Mean Squared Error (MSE) loss between the generated super-resolution image and the ground-truth high-resolution image. This approach optimizes purely for pixel-level reconstruction accuracy. The model is trained with the Adam optimizer at a learning rate of 1×10^{-4} , processing images in batches of 16 over 100,000 iterations. This MSE-based training provides a stable baseline and serves as a pre-trained initialization for SRGAN.

2) *SRGAN*: SRGAN [2] extends the training process by incorporating adversarial training with a discriminator network. The generator is initialized from a pre-trained SRResNet checkpoint to provide a stable starting point. The

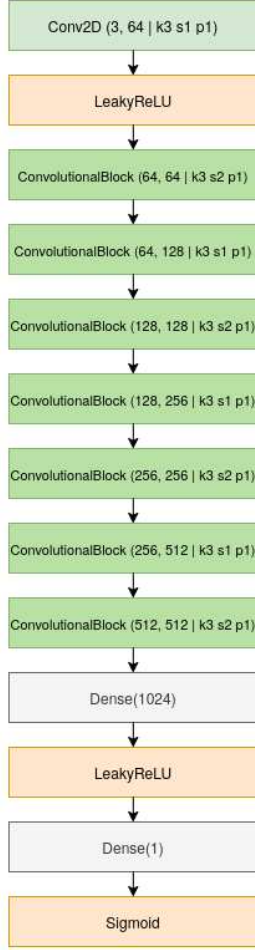


Fig. 5: Discriminator Architecture

training process alternates between generator and discriminator updates. The generator is optimized using a perceptual loss function combining content loss and adversarial loss. Rather than computing pixel-wise MSE, the content loss is calculated using feature representations extracted from a pre-trained VGG19 network [7]. By comparing feature maps from VGG19’s $conv_{5,4}$ layer after activation between generated and ground-truth images, the content loss captures perceptual similarity rather than strict pixel-level correspondence, resulting in sharper and more visually realistic reconstructions [8]. The adversarial loss encourages the generator to produce images that fool the discriminator, weighted at 5×10^{-3} . The discriminator is trained using binary cross-entropy loss with label smoothing (0.1) to prevent overconfidence, comparing its predictions against real (label = 0.9) and fake (label = 0) images. Both networks utilize the Adam optimizer with a learning rate of 1×10^{-4} and are trained for 100,000 iterations.

3) *SRRaResNet*: SRRaResNet modifies the generator architecture by removing all batch normalization layers. The model is trained in PSNR mode without any discriminator, using the combination of pixel (0.01 weight) and perceptual (1.0 weight)

L1 loss. The perceptual loss is computed using the feature maps from VGG19’s $conv_{5,4}$ layer, with this one before activation, similar to [6]. The removal of batch normalization aims to minimize instability, as batch statistics can sometimes interfere with the reconstruction of the image by introducing artifacts. Trained for 100,000 iterations with the same batch size and learning rate as SRResNet, SRRaResNet explores architectural modifications independent of adversarial training and serves as the initialization checkpoint for SRRaGAN.

4) *SRRaGAN*: SRRaGAN represents the most sophisticated training strategy, combining relativistic adversarial training [4] with the architectural modifications from SRRaResNet (no batch normalization in generator, VGG19 without activation). Unlike standard SRGAN which trains the discriminator to classify images as real or fake independently, SRRaGAN employs a relativistic discriminator that estimates the probability that a real image is more realistic than a fake image, rather than judging them in isolation [6]. The generator loss combines three components: perceptual loss computed using VGG19 features (without activation functions) with weight 1.0, pixel-level L1 loss with weight 1×10^{-2} for preserving fine details, and relativistic adversarial loss with weight 5×10^{-3} . The discriminator implements relativistic loss by comparing pairs of real and fake images: it encourages real images to be relatively more realistic than fake ones, while the generator aims to make fake images relatively more realistic than real ones. The model is initialized from the pre-trained SRRaResNet checkpoint and fine-tuned for 100,000 iterations, allowing the adversarial training to refine the pixel-accurate baseline into a perceptually optimized solution.

All models utilize identical infrastructure: batch size of 16, crop size of 96×96 for high-resolution patches, $4 \times$ scaling factor, Adam optimizer with learning rate 1×10^{-4} , 64 channels in residual blocks, and 16 residual blocks. The progressive training strategy (SRResNet $\rightarrow$ SRGAN and SRRaResNet $\rightarrow$ SRRaGAN) allows adversarial models to build upon stable pixel-accurate baselines, improving convergence and final quality.

V. EVALUATION METRICS

Model performance was evaluated using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) on the test dataset. PSNR measures the ratio between maximum pixel value and reconstruction error (higher is better), while SSIM assesses perceptual quality by comparing luminance, contrast, and structure (closer to 1 is better).

VI. RESULT

A. Quantitative Metrics

1) *PSNR-Based Models*: SRResNet achieved the highest quantitative scores (30.89 dB PSNR, 0.8915 SSIM), establishing the performance ceiling for pixel-accurate reconstruction. SRRaResNet, which excludes batch normalization, achieved comparable results (30.37 dB PSNR, 0.8831 SSIM), demonstrating that batch normalization removal does not significantly

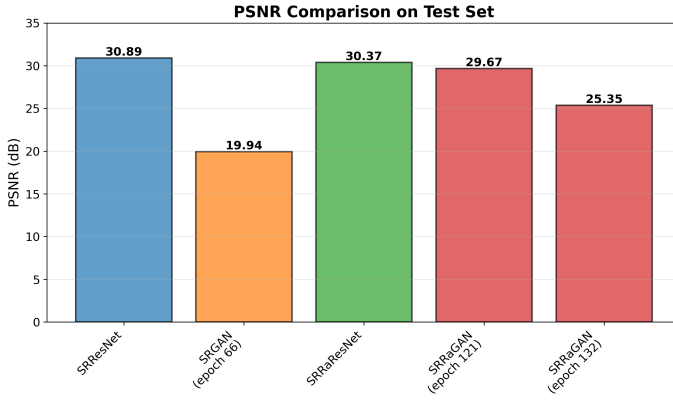


Fig. 6: PSNR Comparison

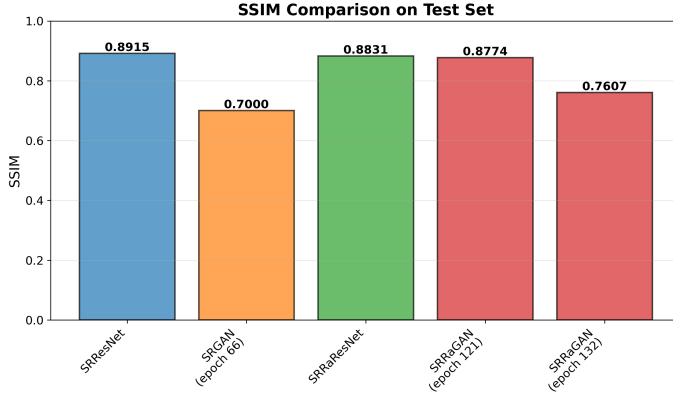


Fig. 7: SSIM Comparison

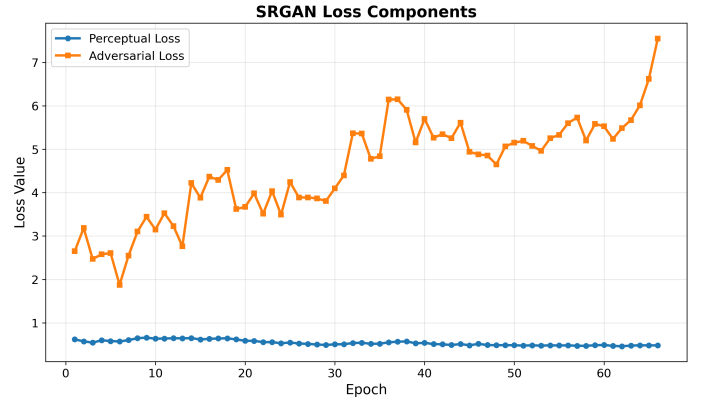


Fig. 8: SRGAN Loss Components

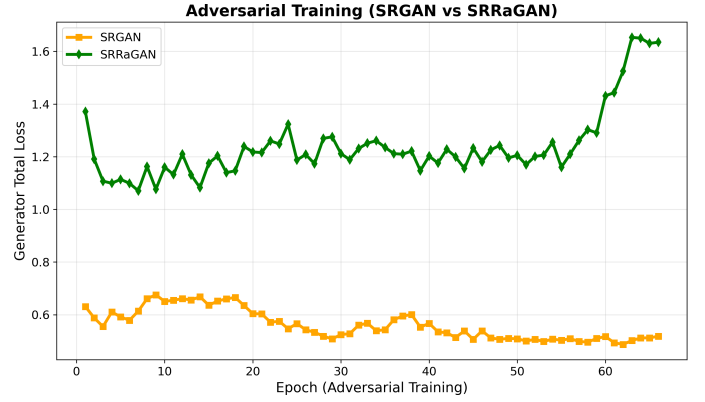


Fig. 9: GAN Models Loss

degrade pixel-level accuracy while potentially preserving fine details critical for anime imagery.

2) *SRGAN Training Instability*: SRGAN exhibited severe training instability, achieving only 19.94 dB PSNR and 0.7000 SSIM at epoch 66. Analysis of the training loss curves reveals erratic behavior and lack of convergence throughout training. This instability likely stems from the purely perceptual loss formulation, where the model optimizes for VGG19 feature matching without sufficient pixel-level constraints. The discriminator may have provided inconsistent gradient signals, causing the generator to converge to a local minimum where perceptual loss appears acceptable but overall image quality is severely degraded. This confirms the motivation for exploring more stable training approaches.

Inspecting the loss graph in Fig. 9 shows that it is consistently going down, even though the adversarial loss is going up as shown in Fig. 8. This suggests that the model is focusing mostly on the perceptual loss and the discriminator isn't punishing enough.

3) *SRRaGAN Performance*: SRRaGAN at epoch 55 achieved 29.67 dB PSNR and 0.8774 SSIM, representing a slight decrease from the PSNR-based baselines. However, this reduction in quantitative metrics is expected and does not necessarily indicate inferior performance. Unlike SRResNet

and SRRaResNet, which optimize purely for pixel-wise MSE, SRRaGAN incorporates perceptual loss components that prioritize feature-level similarity over exact pixel matching. The inclusion of perceptual loss (weight 1.0), pixel-level L1 loss (weight 0.01), and relativistic adversarial loss (weight 0.005) enables the model to generate visually sharper results with enhanced edge definition, even though PSNR and SSIM metrics are designed for pixel-accurate reconstruction, there is the possibility of not fully capture these perceptual improvements. This trade-off between quantitative metrics and perceptual quality is well-documented in the super-resolution literature [2] [6].

4) *Instability in SRRaGAN*: At epoch 66, SRRaGAN's performance degraded significantly to 25.35 dB PSNR and 0.7607 SSIM, suggesting that the model has started collapsing. Extended adversarial training beyond epoch 55 may have caused the generator to overly prioritize discriminator fooling at the expense of reconstruction fidelity, indicating that epoch 55 represents the optimal checkpoint for this model. The behaviour can be observed in Fig. 9, in which the loss for SRRaGAN started going up around epoch 55.

5) *SRGAN vs SRRaGAN*: SRRaGAN demonstrates significantly more stable training compared to SRGAN as shown in Fig. 10, with the generator total loss showing gradual improve-



Fig. 10: All Generator Training Loss

ment before signs of instability become apparent after epoch 55. The relativistic adversarial loss formulation and inclusion of pixel-level L1 loss provide the necessary constraints to prevent complete mode collapse.

B. Qualitative Comparison

1) *SRGAN*: Section VI-A indicated that SRGAN’s training collapsed, yielding a much lower performance compared to the other models. Fig. 11 shows an example output from checkpoints at epoch 40 and 66. The training never yielded a good and stable checkpoint, thus most of the stored checkpoints shows an output with wild colors. The texture of the output image doesn’t seem to be much sharper than it’s SRResNet counterpart, which is investigated in Section VI-B3.



(a) High resolution (b) Epoch 40 output (PSNR: 19.0528 — SSIM: 0.7470) (c) Epoch 66 output (PSNR: 20.1282 — SSIM: 0.7423)

Fig. 11: SRGAN outputs showing extreme colors

2) *SRRaGAN*: As mentioned in Section VI-A4, SRRaGAN became unstable nearing the end of the training, thus prompting us to use a stabler model from previous epochs. Fig. 12 shows the difference between the two checkpoints, in which epoch 66 yields an image that has a tint of green around the edges. This effect is applied across various images.

3) *Texture Quality*: Fig. 13 shows a collage of the model outputs. For the most part, the GAN version of the models are not much better than their non-GAN counterpart. SRRaGAN



(a) Epoch 55 output (PSNR: 25.7187 — SSIM: 0.9363) (b) Epoch 66 output (PSNR: 22.1008 — SSIM: 0.8476)

Fig. 12: SRRaGAN outputs from different epochs

does consistently output a slightly sharper version than SRResNet, which can be observed by looking at specific parts that has complex texture like the eye or the hair. The effect of the batch normalization that was mentioned in [6] can also be observed in one of the output by SRResNet.

VII. DISCUSSION

A. SRGAN Failure Analysis

SRGAN outputs at both epoch 40 and epoch 66 exhibit severe color distortion, with the character rendered in unnatural orange/yellow tones completely different from the ground truth’s blue/purple coloring as seen in Fig. 11. This failure demonstrates complete mode collapse, where the generator produces visually incoherent results despite the discriminator’s feedback. The abnormal coloring artifacts confirm that SRGAN failed to learn meaningful feature mappings, resulting the extremely low quantitative metrics. The model appears to have converged to a degenerate solution where it generates arbitrary color shifts while maintaining basic structural outlines, fooling the discriminator without producing usable super-resolution results.

B. PSNR-Based Models (SRResNet and SRRaResNet)

SRResNet (epoch 66) and SRRaResNet (epoch 66) both produce visually accurate reconstructions that closely match the ground truth coloring and overall appearance. The images maintain proper color fidelity with the characteristic blue/purple tones preserved. Both models demonstrate clean reconstruction of anime-specific features (hair strands definition, eyes details and overall colour). The visual similarity between these two models confirms that batch normalization removal does not introduce perceptible quality degradation, supporting the quantitative findings.

C. SRRaGAN Performance

SRRaGAN at epoch 55 produces results visually comparable to the PSNR-based models, maintaining accurate colors and detail preservation. Notably, despite the slightly

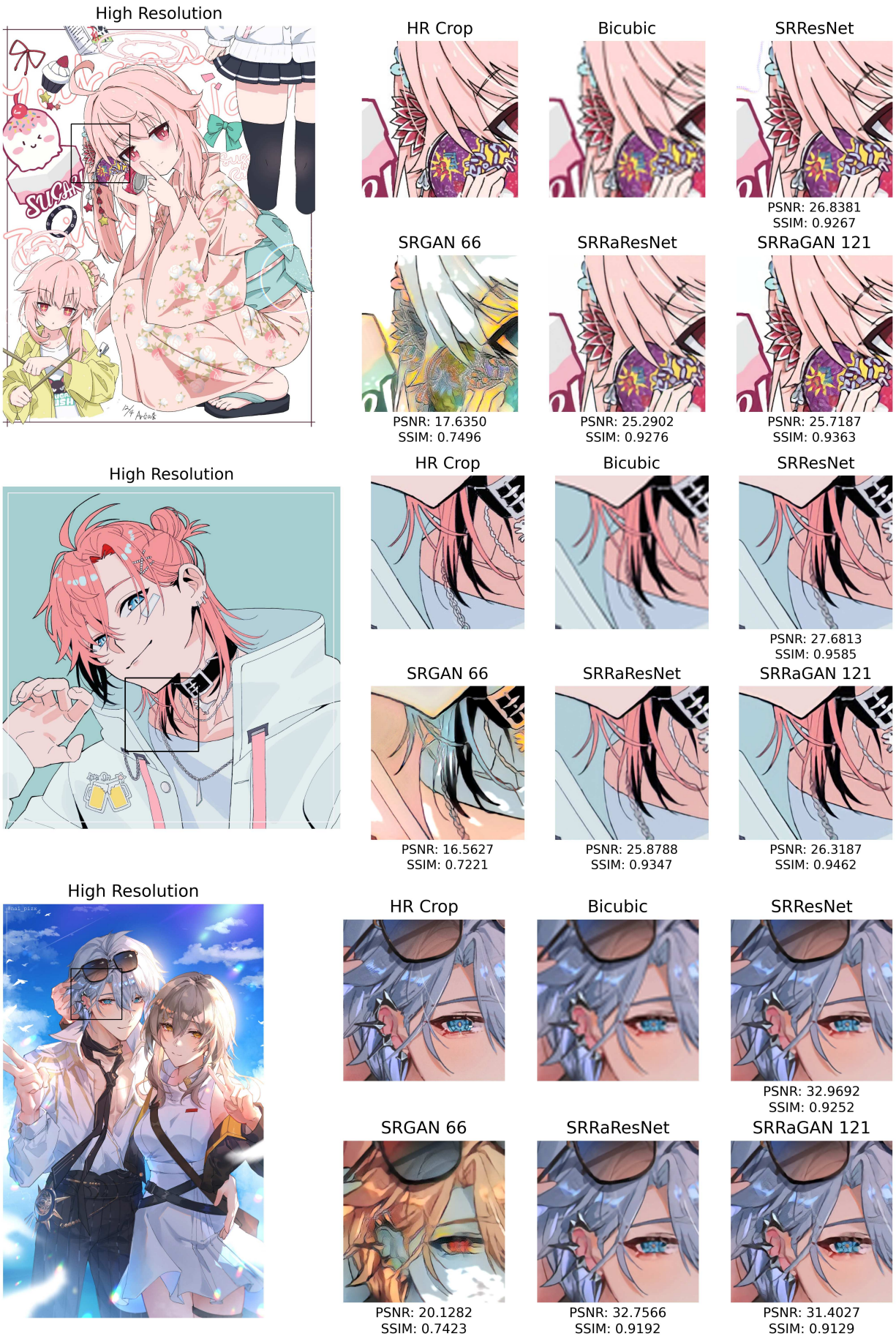


Fig. 13: Model outputs showing textures of certain parts

lower PSNR compared to SRResNet, the perceptual quality appears equivalent or superior in edge sharpness and fine detail rendering. Close examination of hair strands and facial features reveals that SRRaGAN may provide enhanced edge definition compared to pure PSNR optimization, validating the hypothesis that perceptual loss can improve subjective quality even with reduced pixel-level metrics. At epoch 66, SRRaGAN shows visible degradation with slightly darker tones, reduced contrast, and visible artifacts in the form of green coloring of edges. This confirms the start of the collapse observed in Section VI-A, where extended adversarial training began sacrificing reconstruction fidelity.

D. Comparison with Bicubic Baseline

All successful models (SRResNet, SRRaResNet, SRRaGAN epoch 55) substantially outperform bicubic interpolation. The bicubic result exhibits characteristic blurriness, loss of fine details, and softened edges typical of traditional upscaling methods. The deep learning approaches demonstrate clear superiority in preserving anime-specific visual features.

VIII. CONCLUSION

SRRaGAN demonstrated substantially greater training stability than SRGAN, maintaining stable convergence until approximately epoch 55. It also produced sharper reconstructions than SRRaResNet in regions containing complex textures such as hair strands and iris details. Visual artifacts associated with batch normalization were observed in SRResNet outputs, supporting the effectiveness of removing batch normalization layers. The success of the modified SRRaGAN architecture reinforces the role of batch normalization removal and relativistic adversarial loss in stabilizing adversarial super-resolution training.

This study has several limitations. First, SRGAN failed to achieve stable convergence, preventing a direct performance comparison under optimal conditions. Second, the low-resolution inputs were generated using bicubic downsampling, which does not accurately reflect real-world degradation processes such as noise, blur, and compression artifacts.

Future work should focus on incorporating more realistic degradation models when generating low-resolution training data. Modeling real-world image degradation, including sensor noise and compression effects, may improve the generalization of super-resolution models to practical applications.

IX. DEPLOYMENT

The application uses React as the front-end library, in which it is hosted on GitHub Pages. There isn't any back-end worker service that runs the inference, rather, the inference is done by the client itself using the ONNX runtime. To use the ONNX runtime, the model has to be exported into the ONNX format beforehand. A live version may be viewed at <https://erik-san.dev/aol-dl>.

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